

Notes: Kaustia & Knupfer (JF, 2008)
Do Investors Overweight Personal Experience? Evidence from IPO Subscriptions

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1 Big picture

- **Question.** Do investors overweight their own personal investment experiences when deciding whether to participate again in the same financial product?
- **Setting.** The paper studies retail subscriptions to Finnish IPOs from 1995 to 2000 using investor-level registry data and IPO allocation records.
- **Hypothesis.** Reinforcement learning predicts that personally experienced IPO returns should be incrementally positive determinants of future IPO subscriptions.
- **Main result.** Investors who personally experience higher IPO initial returns are much more likely to subscribe to later IPOs.
- **Two-period evidence.** In the main logit, past mean IPO return has a coefficient around 4.65 after controls; a one-standard-deviation increase in personal returns raises future participation probability by about 7.7 percentage points.
- **Event-by-event evidence.** Using the previous three IPO experiences, higher returns predict both participation in the next IPO and a shorter time until the next subscription.
- **Natural-experiment evidence.** Investors whose first IPO experience is positive are more than twice as likely to participate immediately again, and the gap persists across later offerings.
- **Interpretation.** The evidence is consistent with investors learning from personal outcomes in a reinforcement-learning way rather than using only publicly observed IPO-cycle information.
- **Takeaway.** Personal investment experience can generate investor sentiment, repeat participation, and popular investment styles.

2 Incremental contribution of the paper

- **Field evidence on reinforcement learning.** Earlier evidence on reinforcement learning came largely from experiments. This paper studies high-stakes real investment decisions by more than 183,000 retail investors.
- **Personal experience versus public information.** IPOs are public events, but the key variable is the return personally experienced through allocation. Offering fixed effects and IPO-cycle controls help separate personal payoff history from market-wide IPO performance.
- **Sequential experimentation setting.** IPO subscription decisions are discrete, repeated, and linked to clearly measured outcomes. Investors do not need to sell to realize the first-day return measure, so the paper avoids disposition-effect selection.
- **Quasi-natural experiment.** Overlapping hot and cold IPOs allow the authors to compare investors with identical IPO histories who first enter through different return experiences.
- **Primacy of first financial experience.** A first positive IPO experience changes later market participation, linking behavioral finance with marketing ideas about initial product experience.
- **Microfoundation for sentiment.** The paper gives a microfoundation for why hot IPO markets can attract retail participation and why investment styles can become popular after cohorts of investors have favorable experiences.

3 Data and measurement

3.1 Data source and sample

- The data come from the Finnish Central Securities Depository, which records shareholdings, transactions, and IPO allocations by unique investor code.
- The sample covers 57 Finnish IPOs from January 1995 through December 2000.
- The investor sample includes 183,919 retail investors who participate in at least one IPO during the sample period.
- For almost all IPOs, every investor who bid for shares received an allocation. This makes allocations a reliable measure of actual participation.
- The data link each investor to age, gender, portfolio value, and stock-market trading activity.

Interpretation: The data are unusually strong for studying personal experience because the authors observe both the investor’s action and the payoff the investor personally receives.

3.2 Key variables

- **IPO participation:** whether an investor subscribes to a future IPO or to the next IPO.
- **Past mean initial return:** allocation-weighted average first-day IPO return personally experienced by the investor.
- **Past subscription activity:** number of past IPO subscriptions, used to control for persistent investor activity.
- **Portfolio controls:** portfolio value, trading activity, age, age squared, and gender.
- **Investor type measures:** benchmark-adjusted portfolio performance and total portfolio return, used to test private-information and wealth interpretations.

4 Empirical designs and specifications

4.1 Two-period logit

The first design splits the sample into a first half and a second half. The baseline logit is

$$\Pr(\text{Subscription}_i^{2nd} = 1) = \Lambda(\alpha + \beta \text{PastReturn}_i + \gamma \text{PastActivity}_i + \delta' \text{Controls}_i).$$

Interpretation: The coefficient β asks whether investors with better personally experienced first-half IPO returns are more likely to subscribe to at least one second-half IPO, controlling for how active and wealthy they already are.

4.2 Event-by-event next-IPO model

The second design conditions on each investor-offering observation and asks whether returns from the previous three IPO subscriptions predict participation in the next IPO. The paper estimates both a logit and a duration model.

Interpretation: Offering fixed effects absorb common IPO characteristics, market-wide IPO-cycle conditions, and recent IPO returns not personally experienced by the investor. This brings the design closer to isolating personal return experience.

4.3 Natural experiment with paired IPOs

The authors identify six pairs of IPOs with overlapping subscription periods where one IPO has a positive initial return and the other has a negative initial return. They select investors whose first IPO investment is in one of these two IPOs, and who subscribe to only one IPO in the pair. They then compare subsequent IPO participation between investors with a first positive experience and investors with a first negative experience.

Interpretation: This design is powerful because investors begin with the same IPO history: none. Differences in later behavior are therefore easier to attribute to the first personal outcome.

4.4 Alternative explanations

The paper checks private information, general stock-market skill, portfolio rebalancing, wealth effects, preferential treatment, liquidity constraints, and hot-market timing. The core return-experience result survives these checks.

5 Tables and figures

5.1 Table I and Figure 1: sample structure and transition probabilities

Read: Table I shows 57 IPOs, average underpricing of 22.27%, and 183,919 investors. The median investor subscribes only once, while the mean is 2.04 subscriptions. Figure 1 shows that investors with poor first-half IPO return experiences are much more likely to make no second-half subscriptions.

Interpretation: Table I establishes that the setting combines volatile IPO payoffs with a large set of retail investors. Figure 1 shows the behavioral pattern before regressions: bad personal experience induces exit, while good experience induces continuation.

Table I
Descriptive Statistics

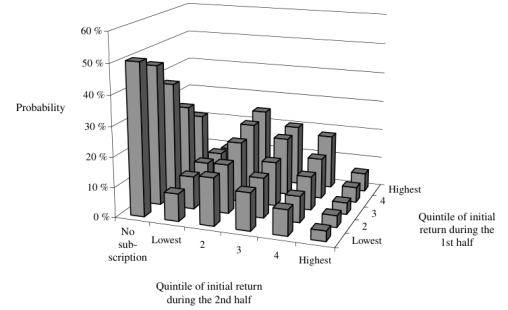
This table gives descriptive statistics of the IPOs and investors in the sample. Panel A shows statistics for Finnish IPOs from 1995 to 2000. Underpricing is calculated as the difference between the first-day closing price and offer price divided by the offer price. Number of investors equals the number of retail investors who were allocated shares in the offering and gross proceeds equals the number of allocated primary and secondary shares to retail and institutional investors times the offer price, before the use of any green shoe options. Panel B shows averages of the same variables and the number of IPOs for each year during the sample period. Panel C shows statistics for investors who participate in at least one IPO during the sample period. Allocation value equals the number of shares allocated to an investor times the subscription price. Portfolio value is calculated as the sum of the value of holdings in individual shares and number of shares as the number of individual shares in the portfolio. Portfolio characteristics are calculated for each investor in the beginning of the subscription period of each offering starting from the first IPO investment, averaged across offerings, and then across investors. Trading activity variables and age come from the beginning of the subscription period of the last IPO in the sample.

Panel A: IPOs, $N = 57$					
Variable	Mean	Median	SD	25% Percentile	75% Percentile
Underpricing	22.27%	2.80%	53.51%	0.00%	20.43%
Number of investors, Th.	7.78	1.97	15.37	1.16	5.24
Gross proceeds, Mill. euros	86.29	27.18	200.03	10.21	0.08

Panel B: IPOs by Year, $N = 57$						
	1995	1996	1997	1998	1999	2000
Average underpricing	0.77%	17.80%	14.41%	14.41%	35.83%	21.23%
Average number of investors, Th.	5.10	0.31	1.96	18.98	7.12	7.62
Average gross proceeds, Mill. euros	77.82	13.21	53.79	311.62	49.32	42.84
Number of IPOs	5	2	8	8	19	15

Panel C: Investors, $N = 183,919$					
	Mean	Median	SD	25% Percentile	75% Percentile
No. of IPOs after the first subscription	32.88	37.00	13.32	18.00	38.00
Number of IPO subscriptions	2.04	1.00	2.47	1.00	3.00
Total value of allocations, Th. euros	3.86	1.35	66.38	0.19	4.64
Value of allocation, Th. euros	1.72	1.13	33.16	0.52	2.64
Portfolio value, Th. euros	64.67	9.20	1,096.12	2.72	29.10
Number of shares in portfolio	3.60	2.57	3.65	1.18	4.67
Value of stock market buys, Th. euros	133.29	7.57	1,976.98	0.50	37.32
Value of stock market sells, Th. euros	152.34	11.89	2,054.75	2.34	49.61
Age	46.34	47.00	17.84	33.00	59.00
Gender dummy	0.66	1.00	0.47	0.00	1.00

(a) Table I: Descriptive statistics



(b) Figure 1: Transition probabilities

5.2 Table II and Figure 2: two-period regression evidence

Read: Table II shows that past mean initial return is positive and highly significant in all specifications. With full controls, the coefficient is 4.65 and the one-standard-deviation probability effect is 7.7 percentage points. Figure 2 shows the same pattern nonparametrically across return-by-activity categories.

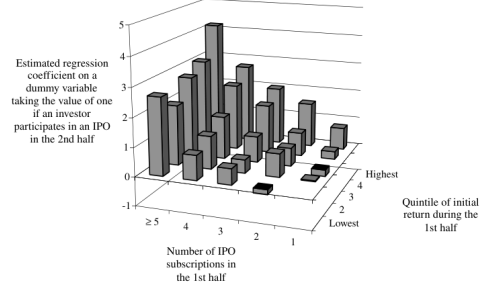
Interpretation: The return effect survives controls for activity, wealth, trading, age, and gender. Figure 2 shows that the effect is not merely persistence in activity: within activity categories, better personal returns lead to higher future participation.

Table II
Two-Period Analysis of the Relation between Past Returns
and Future IPO Subscriptions

The sample period is divided into two halves. The first half has 40 IPOs, the last occurring in October 22, 1999; the second half has 17 offerings, the last in December 20, 2000. This split is determined by placing an equal number of investor/offering pairs in both periods. We include investors who participate in at least three IPOs in the first half. Regressions are run using a logit model. The dependent variable is one if an investor participates in at least one IPO in the second half, zero otherwise. Past mean initial return is the average initial return from the IPOs in which the investor participates, weighted by the value of allocation. Unreported control variables are age, age squared, a dummy for being male, quintile dummies for portfolio value, and quintile dummies for value of trades (summed from the beginning of the sample period). Columns 1 through 4 report coefficient estimates of regressions. Column 5 reports the changes in probabilities associated with a one *SD* increase in an explanatory variable, evaluated at the mean of all other variables. Probability of reference investor is the model's predicted probability to participate, evaluated at the mean of all the explanatory variables; the male dummy set to zero; and the omitted quintile dummy set to zero for both portfolio value and value of trades. *t*-values (in parentheses below coefficients) are based on robust standard errors.

Specification	Logit (0 = No Future Subscriptions, 1 = At Least One Future Subscription)				Reference Probability and Probability Changes
	1	2	3	4	
Probability of reference investor					70.4%
Past mean initial return	5.41 (25.50)	4.67 (21.68)	4.67 (21.72)	4.65 (21.56)	7.7%
Number of past IPO subscriptions	0.55 (37.56)	0.47 (32.29)	0.47 (32.29)	0.47 (32.14)	24.0%
Portfolio value dummies	No	Yes	No	Yes	
Trading activity dummies	No	No	Yes	Yes	
Pseudo- <i>R</i> ²	0.21	0.24	0.24	0.24	
Number of observations	21,928	21,928	21,928	21,928	

(a) Table II: Two-period logit



(b) Figure 2: Participation by activity and returns

5.3 Table III: next-IPO participation and duration

Read: Past mean initial return predicts the decision to subscribe to the next IPO: coefficient 0.47 with t-statistic 18.78. The probability change is 6.90%. In the duration model, the coefficient is 0.43 with a 33.70% hazard-ratio change.

Interpretation: Table III is stronger than the two-period design because offering fixed effects absorb common offer-level shocks. Conditional on the same IPO opportunity, investors with better personal histories are more likely to come back and come back sooner.

5.4 Table IV: natural experiment based on first IPO experience

Read: After the first subsequent offering, investors with a positive first IPO return participate at 10%, while investors with a negative first IPO return participate at 4%. By the tenth subsequent offering, the cumulative rates are 65% versus 39%. For stock-market entrants via IPO, the tenth-offering rates are 67% versus 38%.

Interpretation: Table IV is the paper's cleanest behavioral evidence. These investors have no prior IPO experience, and the initial hot-versus-cold experience has persistent effects. The persistence suggests durable updating of taste, confidence, or willingness to use the IPO product.

5.5 Table V: investor type, skill, and alternative explanations

Read: Past mean initial return remains strongly positive, around 4.37 to 4.48. Interactions with portfolio performance and portfolio return are statistically insignificant. The portfolio return indicator itself is positive, but it does not absorb the IPO-specific personal return effect.

Interpretation: Table V weakens private-information and wealth interpretations. If the result were mainly skill, top performers should react differently from bottom performers. If it were wealth, total portfolio returns should absorb the effect. Neither happens.

Table III

Timing of Participation to Next IPO as a Function of Past Returns

This table gives results of investor-level regressions where the dependent variable is either an indicator variable representing the decision to subscribe to the next IPO (column 1 of Panel A) or the time to the next IPO subscription (column 2 of Panel A). For every investor participating in at least three IPOs during the sample period from 1995 to 2000, the time to the next IPO subscription is determined by recording how many IPOs elapsed until the investor participates in the next IPO, including the next IPO. The dependent variable is regressed on a set of explanatory variables using a logit model in column 1 and a duration model with an exponential duration distribution in column 2. The hazard ratio reported in column 2 is given by $\exp$ (coefficient obtained from the duration regression). The independent variable is the average initial return from the previous three IPOs subscribed. Unreported control variables are age, age squared, a dummy for being male, quintile dummies for portfolio value, quintile dummies for value of trades (summed over the previous 120 trading days), dummies for the past number of subscriptions (from three to ten or more), and dummies for each offering. "Probability change" shows the changes in probabilities associated with a one SD increase in an explanatory variable, evaluated at the mean of all other variables. "Hazard ratio change" is the change in hazard ratio associated with a one SD increase in an explanatory variable. t -values (in parentheses below coefficients) are based on robust standard errors.

Panel A: Regression Results				
Specification	1		2	
	Decision to Subscribe to Next IPO		Time to Next Subscribed IPO	
Dependent Variable	Coefficient	Probability Change	Hazard Ratio	Hazard Ratio Change
Past mean initial return	0.47 (18.78)	6.90%	0.43 (24.32)	33.70%
Dummies for Offerings		Yes		Yes
Portfolio value quintiles		Yes		Yes
Trading activity quintiles		Yes		Yes
Subscription activity		Yes		Yes
Number of observations	110,501		115,040	
Panel B: Descriptive Statistics				
	All Observations		At Least Three Subscriptions	
Number of observations	321,394		115,040	
Number of failures	191,195		77,457	
Time to next IPO, mean	12.61		7.26	
Time to next IPO, median	11		3	

Table III: Timing of participation to next IPO

Table IV reports the results. The first two columns show the proportion of first-time IPO investors participating in further IPOs, conditional on having

Table IV

Natural Experiment of Survival to Next IPO

This table gives the proportion of investors participating in a subsequent IPO conditional on having experienced either a negative or a positive initial return in their first IPO. The sample includes six pairs of IPOs. To qualify as a pair in this test, the two IPOs must have simultaneously running subscription periods, and it must be the case that one IPO has a positive initial return, while the other one has a negative initial return. The sample includes all such pairs. The "cumulative proportion of participating investors" is given by the number of investors who have made at least one further subscription since the initial IPO divided by all investors participating in the initial IPO, and it is calculated for ten subsequent offerings. "First IPO" includes investors who make their first IPO investment in either one of the IPOs in a matched pair. "Stock market entry via IPO" includes only investors who have made neither IPO nor other common stock investments before. The z -values are for a test of difference in proportions.

Cumulative Proportion of Participating Investors								
Number of Offering after the First IPO with Positive or Negative Initial Return	First IPO				Stock Market Entry Via IPO			
	Pos. Initial Return	Neg. Initial Return	Difference Between Proportions	z -Value	Pos. Initial Return	Neg. Initial Return	Difference Between Proportions	z -Value
1 st	0.10	0.04	0.06	7.7	0.11	0.07	0.05	2.1
2 nd	0.20	0.07	0.13	12.6	0.24	0.09	0.14	5.0
3 rd	0.33	0.13	0.20	15.3	0.38	0.16	0.22	6.0
4 th	0.43	0.17	0.26	17.5	0.50	0.19	0.30	7.3
5 th	0.45	0.18	0.28	17.7	0.53	0.21	0.32	7.1
6 th	0.50	0.19	0.30	18.7	0.55	0.22	0.33	7.2
7 th	0.56	0.21	0.35	20.9	0.58	0.24	0.34	7.1
8 th	0.60	0.35	0.25	12.9	0.63	0.34	0.29	5.8
9 th	0.64	0.37	0.28	13.0	0.67	0.35	0.32	5.7
10 th	0.65	0.39	0.26	11.0	0.67	0.38	0.30	4.7
Number of investors at the beginning of	2,304	2,182			412	252		

Table IV: Natural experiment of survival to next IPO

Table V

Effect of Past Return for Different Investor Types

This table reports regressions similar to that in Table II column 4 except that the past return variable is interacted with two indicator variables. Investors are first divided into quintiles based on performance (measured as own benchmark adjusted returns, see text) and total portfolio return, both calculated for the first half of the sample. Investors with no trading activity during the first half are left out of the analysis. For every investor, indicator variables for performance and portfolio return are defined in the following way. If an investor belongs to the highest quintile, the indicator gets a value of one. If she belongs to the lowest quintile, the indicator takes a value of -1 . All other investors are given the value of zero. The t -values (in parentheses below coefficients) are based on robust standard errors.

Specification	Logit (0 = No Future Subscriptions, 1 = At Least One Future Subscription)		
	1	2	3
Past mean initial return	4.48 (19.79)	4.37 (19.23)	4.38 (19.25)
× Performance indicator	−0.22 (−0.64)		−0.18 (−0.52)
× Portfolio return indicator		−0.22 (−0.60)	−0.22 (−0.59)
Performance indicator	0.04 (0.55)		0.02 (0.35)
Portfolio return indicator		0.24 (3.43)	0.24 (3.41)
Number of past IPO subscriptions	0.45 (30.60)	0.46 (30.65)	0.46 (30.65)
Portfolio value dummies	Yes	Yes	Yes
Trading activity dummies	Yes	Yes	Yes
Pseudo- R^2	0.21	0.21	0.21
Number of observations	19,039	19,039	19,039

Table V: Effect of past return for different investor types

6 Short summary

- The paper studies whether retail investors overweight personal IPO experience when deciding whether to subscribe again.
- The setting is Finnish IPOs from 1995 to 2000, using registry data that link each investor to IPO allocations, returns, and later actions.
- Past IPO return experience strongly predicts future IPO participation.
- A one-standard-deviation increase in past IPO return raises future participation probability by 7.7 percentage points in the two-period design.
- In event-by-event regressions, better recent IPO experience predicts a 6.9 percentage point increase in next-IPO participation and a 33.7% higher participation hazard.
- The overlapping IPO-pair natural experiment shows that first hot IPO experiences have persistent consequences for future participation.
- The effect is not explained by private information, portfolio performance, wealth effects, liquidity constraints, or the IPO cycle.

7 Conclusion

7.1 Core conclusion

Kaustia and Knupfer show that personal investment experience is a powerful determinant of future investor behavior. Retail investors who personally earn high returns in IPOs are more likely to return to the IPO market, while investors who personally experience poor IPO returns are more likely to stop participating.

7.2 Economic intuition

A good IPO return observed in the newspaper is not psychologically equivalent to a good IPO return deposited into the investor's own account. Personal payoff experience changes confidence, salience, memory, and willingness to repeat the action. This creates reinforcement learning: good outcomes increase repetition, while bad outcomes reduce it.

7.3 Contribution to behavioral finance

The paper turns reinforcement learning from an experimental concept into a field mechanism in financial markets. It also gives a microfoundation for IPO sentiment: cohorts of investors who personally experience high IPO returns can become more willing to participate in subsequent offerings.

7.4 Implications for IPO markets and financial products

Hot IPO markets can generate continuing retail demand because positive initial experiences encourage repeat participation. Cold IPO outcomes can reduce the future retail investor base. More broadly, early customer experience can matter for adoption of financial products such as IPOs, mutual funds, structured products, and trading platforms.

7.5 Limits and future research

The setting is Finland during a specific period with discrete IPO decisions, so external validity should be tested in other markets and products. Future work could link experience to measured beliefs, study spillovers

across product categories, and estimate how cohorts of personal experience aggregate into market-wide sentiment.

7.6 Takeaway

Investors learn from what happens to them, not only from what they observe. Personal payoff history is therefore a state variable in financial participation, sentiment, and asset-demand dynamics.